Graphics Library

The following routines are available in this library:

Name	Address	Description	Inputs	Outputs
gr_pixel	0x3000	Set a pixel with colour ‘col’ at (x,y)	R0: x R1: y R2: col	-
gr_line	0x31a0	Draw a line with colour ‘col’ from (x1,y1) to (x2,y2)	R0: x1 R1: y1 R2: x2 R3: y2 R4: col	-
gr_rect	0x33f0	Draw a rectangle with colour ‘col’ and corners (x1,y1) & (x2,y2)	R0: x1 R1: y1 R2: x2 R3: y2 R4: col	-
gr_rect_fill	0x3450	Draw a filled rectangle with colour ‘col’ and corners (x1,y1) & (x2,y2)	R0: x1 R1: y1 R2: x2 R3: y2 R4: col	-
gr_circle	0x34c0	Draw a circle with colour ‘col’, centre (x,y) and radius ‘r’	R0: x R1:y R2: r R3: col	-
gr_circle_fill	0x37d0	Draw a filled circle with colour ‘col’, centre (x,y) and radius ‘r’	R0: x R1:y R2: r R3: col	-
gr_fp_pixel	0x3a40	Set a pixel with colour ‘col’ at (x,y), where x & y are floating point numbers	R0: x R1: y R2: col	-

Any programs using these routines should include the following definitions:

```
.equ gr_pixel       0x3000
.equ gr_line        0x31a0
.equ gr_rect        0x33f0
.equ gr_rect_fill   0x3450
.equ gr_circle      0x34c0
.equ gr_circle_fill 0x37d0
.equ gr_fp_pixel    0x3a40
```

And some colour definitions:

```
.equ GR_RED       0xf800
.equ GR_BLUE      0x001f
.equ GR_GREEN     0x07e0
.equ GR_DK_GREEN  0x0560
.equ GR_BLACK     0x0000
.equ GR_WHITE     0xffff
.equ GR_BROWN     0x9260
.equ GR_LT_BROWN  0xdd0d
.equ GR_GOLD      0xfea0
.equ GR_SILVER    0xbdf7
.equ GR_ORANGE    0xfcc0
.equ GR_TURQUOISE 0x273a
.equ GR_PURPLE    0xa01f
.equ GR_PINK      0xebda
.equ GR_FUCHSIA   0xf81f
.equ GR_CYAN      0x07ff
.equ GR_BKGRD     0xffd7
```

When using these routines in ‘ASM’ code the ‘Input Registers’ should be set up, then a ‘call func’ instruction used, with any result then being available in R0 following the ‘call’.

And when using ‘RMG’ code the ‘Input Registers’ are used as arguments to the function call, with any result being assigned as required.

Here are ‘ASM’ and ‘RMG’ examples of ‘gr_rect’ being used:

```
; Draw a brown rectangle with corners (55,55) & (60,60)
ldi r0,55
ldi r1,55
ldi r2,60
ldi r3,60
ldi r4,GR_BROWN
call gr_rect
```

```
// Draw a brown rectangle with corners (55,55) & (60,60)
gr_rect(55, 55, 60, 60, GR_BROWN);
```